

Summary Comparison

Program	Collective(s) used	Array allocation	Manual summation loop at root?	Where final result is available
sum_bcast	MPI_Bcast	Every process allocates the full array	Yes — root must manually add the local sums	Root only
sum_scatter	MPI_Scatter	Each process receives only a chunk of the array	Yes — root manually adds all local sums	Root only
sum_gather	MPI_Scatter + MPI_Gather	Each process receives only a chunk	Yes — root manually sums the gathered local sums	Root only
sum_reduce	MPI_Scatter + MPI_Reduce	Each process receives only a chunk	No — MPI_Reduce performs the summation	Root only
sum_allreduce	MPI_Scatter + MPI_Allreduce	Each process receives only a chunk	No — MPI_Allreduce performs the summation	All processes
sum_scan	MPI_Scatter + MPI_Scan	Each process receives only a chunk	No — MPI_Scan calculates the prefix sums	Different result on each rank

Execution times

Program	2 processes	4 processes	8 processes
sum_bcast	0.0053 s	0.0137 s	0.0149 s
sum_scatter	0.0073 s	0.0073 s	0.0056 s
sum_gather	0.0064 s	0.0207 s	0.0051 s
sum_reduce	0.0063 s	0.0052 s	0.0085 s
sum_allreduce	0.0061 s	0.0072 s	0.0052 s
sum_scan	0.0133 s	0.0124 s	0.0123 s

The execution times varied depending on the number of processes. With 2 processes, `sum_bcast` was fastest at 0.0053 seconds. With 4 processes, `sum_reduce` was fastest at 0.0052 seconds, while with 8 processes, `sum_gather` was fastest at 0.0051 seconds. The differences are small, so the results can vary between executions. Generally, collective reduction operations such as `MPI_Reduce` and `MPI_Allreduce` are useful because MPI handles the communication and summation instead of requiring a manual summation loop.

I would choose `MPI_Scan` over `MPI_Allreduce` when each process needs a cumulative result rather than the same global result. For example, if each MPI process has a number of elements that it needs to place into a shared output array, `MPI_Scan` can calculate the cumulative offset for each process. `MPI_Allreduce` would instead give every process the same overall total.